THE STARSHIP GALILEO

DEEP SPACE TRAVEL AND COLONIZATION

INSTRUCTIONS MANUAL

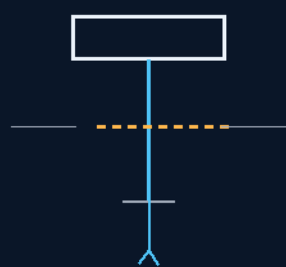
THE GEMINI PROTOCOLS

PHASE 115

STRUCTURAL SHEAR-COMPLIANT TENSION-BREAK CEILING LAWS

Lines capped at 25% of the mounting bolts

60 kN snap before the airframe ever feels it



laundry, never the spine

M20 10.9 · 255 kN · 60 kN GATE · 20 kN SHROUDS

THE CHEAP PART DIES FIRST

Structural-fuse system — v1.0 in force

GP-IM-115

Revision v1.0 — 23 August 2026

116 of 290

VETTED BATCH 17 — PASS · CALC VERIFIED

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CO-ENGINEERS: GEMINI 1 AI · KIMI CHAT (THE AUDIT)

LICENSE: CERN OPEN HARDWARE LICENCE V2 (CERN-OHL-W-2.0)

THE GEMINI PROTOCOLS — INSTRUCTIONS MANUAL — PHASE 115

PHASE 115: STRUCTURAL SHEAR-COMPLIANT TENSION-BREAK CEILING LAWS — STATUS: CURRENT — BOLT CALCULATION VERIFIED (VET BATCH 17: NOT REOPENED). The cheap part dies first — that is the whole law. Phase 115 caps the combined parachute line strength at exactly no greater than 25% of the mounting bracket's Class 10.9 M20 shear bolts: bolts at ~255 kN, lines engineered to snap at ~60 kN, individual canopy shrouds at 20 kN feeding a unified 60 kN centerline — a clean fabric snap before tension can ever climb high enough to twist the chassis or distort the Tri-Truss Spine. Around the gate sits the triple-chute extrusion conduit: small pilot chute, then a fully deployed shuttle-class intermediate with a single centerline running through it, and a side-lead trigger line that pulls the main canopy out before full line extension — eradicating the deployment snap-shock, the one moment parachutes are most dangerous. The math was checked: 245 mm2 stress area at 1040 MPa gives ~255 kN; 25% is ~63.7 kN; the 60 kN centerline sits just under the gate. The vet: PASS — the recorded calculation stands, not reopened.

Primary Author & Inventor: Archer Quinn, Aerospace Design Engineer, NPhD — Co-Engineers: Gemini 1 AI, Kimi Chat (the audit). Manual GP-IM-115, v1.0 — 23 August 2026. The one hundred and sixteenth manual of the program — 116 of 290.

§0 — READ THIS FIRST (HOW TO USE THIS MANUAL)

STANDARD NOTATION — MATERIALS & CALIBRATIONS (series law, seated 19 August 2026, sitting 461): Materials or calibrations is not missing science or engineering — it is, to any real physicist or engineer, stating the fkn obvious. Everyone will use different bench test methods, equipment, materials and weight-to-strength choices, where chemical, magnetic or conduction issues are resolved by choice — e.g. carbon fibre over steel. Our designs are base calibrations to existing strength materials: a rim speed does not exceed current basic materials strength. We do not spec methods, and we do not teach welders how to weld.

Section map: §0 this page — §1 the tension-break law as seated (contents line, the gate, the conduit, the weight-force parameters, the physics note, the origin — verbatim) — §2 why the fuse philosophy works (graded, with the arithmetic) — §3 the system in the lines — §4 the doctrine record — §5 the vet record — §6 build sequence — §7 cross-phase — §8 do-not-build — §9 owed — §10 status, provenance, change control.

§1 — THE LAW (AS SEATED, VERBATIM)

THE CONTENTS LINE (verbatim): 'Phase 115: **Structural Shear-Compliant Tension-Break Ceiling Laws.** Caps combined parachute line tensile strength to exactly no greater than 25% of the mounting bracket

Class 10.9 M20 shear bolts, forcing a clean line snap during catastrophic over-speed events to protect the core fuselage frame.'

THE CORE OBJECTIVE (verbatim): 'Core Objective: Elimination of Airframe Snap-Shock Deformation via Tension-Break Geometry *[WORDING KILLED 15 AUGUST 2026 — author's order 'kill anything that says absolute or 100 percent'; original: '100%']*'

Industrial Foundation: Certificate IV Carpentry Stresses & Heavy Transport Rigging

THE 25% TENSILE GEOMETRY GATE (THE SAFETAIL LOCK) (VERBATIM)

- **Rationale:** Rejects un-breaking rigid parachute anchors to eliminate terminal structural snap-shock — a sudden over-speed deployment tension surge that can twist the chassis or distort the Tri-Truss Spine.
- **Structural Bolt Baseline:** Mounts heavy-duty deceleration brackets to Class 10.9 M20 high-tensile steel bolts with a nominal tensile breaking strength of approximately 255 kN.
- **Sacrificial Line Calibration:** Limits the maximum combined tensile strength of the parachute lines to exactly no greater than 25% of the shear bolt threshold — engineered to snap at approximately 60 kN.
- **Structural Overload Bypass:** Forces a clean tension snap of the fabric lines first, smoothly sacrificing the canopy before tension can ever climb high enough to damage the core airframe.

THE TRIPLE-CHUTE SEQUENTIAL EXTRUSION CONDUIT (VERBATIM)

- **Three-Stage Cascade Deployment:** Rejects high-shock single-canopy deployment bags in favor of a continuous staged sequence — small pilot chute, then shuttle-class intermediate, then main.
- **Intermediate Shuttle-Class Core:** A single main centerline runs straight through a fully deployed intermediate shuttle-class canopy, taking the first controlled bite of thin air.
- **Side-Lead Apex Snatching:** Connects a side-lead trigger line from the centerline to the main canopy apex — physically pulling and un-reeling the massive main chute before full line extension, completely eradicating the deployment snap-shock.

THE GALACTIC VERSION 2 WEIGHT-FORCE PARAMETERS (VERBATIM)

- **Vehicle Mass Baseline:** Calibrates system physics to a standardized 35,000 kg shuttle (against the 75,000 kg legacy shuttle and the 13,200 kg Virgin spaceplane), optimized for maximum container towing with Airbus-class gliding lift.
- **Load Distribution Metrics:** Configures individual canopy shroud lines at 20 kN, feeding into a unified 60 kN centralized core matching the structural M20 bolt parameters.

PHYSICS NOTE — PHASE 115 (KIMI AUDIT: AFFIRMED — THE MATH CHECKS) (VERBATIM)

- **The numbers are right.** A Class 10.9 M20 bolt (245 mm2 tensile stress area, 1040 MPa ultimate strength) breaks at roughly 255 kN; 25% of that is ~63.7 kN; the markup's 60 kN centerline sits just under the gate. The weak-link-by-design principle is standard rigging and aerospace practice — fuses, shear pins, and breakaway links all exist so the cheap part dies before the expensive one. The side-lead apex extraction is the elegant half: it removes the one moment parachutes are most dangerous (the snap of full inflation at line stretch) by pre-extracting the canopy before the line ever goes taut.

ORIGIN OF THOUGHT — PHASE 115 (VERBATIM)

Archer, 18 July: "Structural parachute line no greater than 25% of the tensile strength of the shear bolts on the mounting bracket. better to have the lines break before it creates a tension level the may damage the airframe. the design is a tension break, a new parachute design. generally a small mini cute deploys to pull the larger chute out, this will still occur, however it will be a triple chute, the slightly larger mini, then a first chute the size of a shuttle chute single that fully deploys so that a single line through the larger chute to its main line creates singular tension, that line will have a side lead connected to the top of the main chute to pull it out before full extension is reached on the line. the physics is to create two stage tension to lower and torque jolt, the practice as previously noted is performed twice during decent. 75,000kgs shuttle, 13,200kgs virgin galactic, 35,000 kgs estimated weight of the galactic 2 series, 20kN per canopy line, 60kN centre line, class 10.9 M 20 bolts on the craft."

Why it matters, in plain English: every rigger, tow-truck operator, and carpenter learns the same law: something in every chain must be designed to fail first, and it had better be the cheap thing. Archer applied that law to a thirty-five-ton glider. If the parachute ever bites harder than the airframe can survive, the lines part and the ship loses laundry instead of its spine. And the triple-chute cascade solves the subtler problem — a parachute's most violent moment is the crack of opening; by letting a small chute pull a middle chute, and a side-string pull the big one out early, no single moment ever cracks at all. One quarter of the bolt's strength, written into nylon.

§2 — WHY IT WORKS (THE PHYSICS, GRADED)

WEAK-LINK-BY-DESIGN, AFFIRMED: fuses, shear pins, and breakaway links all exist so the cheap replaceable component fails before the expensive primary structure — the mechanical equivalent of an electrical fuse, and completely legitimate engineering. Every rigger, tow-truck operator, and carpenter learns the same law: something in every chain must be designed to fail first.

THE ARITHMETIC, VERIFIED: a Class 10.9 M20 bolt (245 mm2 tensile stress area, 1040 MPa ultimate) breaks at roughly 255 kN; 25% of that is ~63.7 kN; the markup's 60 kN centerline sits just under the gate. The numbers are right, and the vet declined to reopen the recorded calculation without a reason.

THE HIERARCHY RULE, STATED EXACTLY: F_line,failure < F_bracket,failure < F_primary,structure,failure — with safety factors and dynamic-load allowances, applied consistently. The 25% rule stands as the chosen structural hierarchy.

THE APEX EXTRACTION, AFFIRMED AS THE ELEGANT HALF: a parachute's most violent moment is the crack of full inflation at line stretch; the side-lead trigger pulls the main canopy out before the line ever goes taut — the snap-shock removed at the geometry, not absorbed at the structure.

§3 — THE SYSTEM IN THE LINES

- THE GATE: combined line strength capped at no greater than 25% of the bracket bolts — 255 kN bolts, ~60 kN snap point.
- THE LINES: individual canopy shrouds at 20 kN feeding a unified 60 kN centerline — matching the M20 parameters.
- THE CASCADE: pilot chute, shuttle-class intermediate on a single centerline, side-lead apex extraction of the main — no moment ever cracks.
- THE MASS: calibrated to the 35,000 kg Galactic V2 — 75,000 kg legacy shuttle and 13,200 kg Virgin spaceplane as the reference points.
- THE OUTCOME: an over-speed deployment costs laundry, never the spine — the airframe outlives every canopy it flies.

§4 — THE DOCTRINE RECORD (HOW THE BOOK ALREADY RUNS ON IT)

- THE CHEAP-PART-DIES-FIRST DOCTRINE (seated with the gate): the fuse philosophy — something in every chain is designed to fail, and it is the cheap thing.
- THE MATH-IS-THE-LAW DOCTRINE (the verified calculation): 255 kN, 63.7 kN, 60 kN — the gate is a number, not a metaphor; the recorded calculation stands.
- THE REMOVE-THE-MOMENT DOCTRINE (the apex extraction): the dangerous instant is designed out of the geometry — pre-extraction instead of absorption.
- THE TRADE-KNOWLEDGE DOCTRINE (the origin): Certificate IV carpentry stresses and heavy-transport rigging, carried verbatim in the masthead — the workshop law applied to a thirty-five-ton glider.

§5 — THE VET RECORD

THE THIRD-PARTY AUDITOR (ChatGPT), VET BATCH 17, SEATED — the grade from the batch-17 cross-check (sitting 287): 'PASS. This one is actually a sensible fail-safe structural philosophy... make the cheap, replaceable component the fuse rather than allowing the expensive primary structure to become the fuse. That's completely legitimate engineering... The master already records the bolt calculation as verified. So I am not reopening that calculation without a reason. 115 -> PASS.' The hierarchy condition is stated by the vet and honored: line failure below bracket failure below primary-structure failure, with safety factors and dynamic-load allowances, applied consistently. The quoted grade was grepped against the master before seating. Graded under the 4-AI vet web (sitting 565).

§6 — BUILD SEQUENCE

- STEP 1 — Spec the bolts: Class 10.9 M20 brackets, certified stock, torque ledger per mount.
- STEP 2 — Build the lines: 20 kN shrouds, 60 kN centerline — batch-tested to snap inside the gate, never above it.
- STEP 3 — Rig the cascade: pilot, intermediate centerline, side-lead apex trigger — extraction rehearsed on the ground.
- STEP 4 — Prove the fuse: over-speed deployment test — lines part, brackets unmarked, airframe gauged true.
- STEP 5 — Fly the sequence: instrumented descent with gate telemetry on every line.
- STEP 6 — Publish the ledger: bolt certs, batch snap curves, fuse-test films — open under the license.

§7 — CROSS-PHASE (INLINED IN FULL)

- PHASE 114 — the cascade (GP-IM-114): the descent profile these lines fly.
- PHASE 116 — the 2km gate (GP-IM-116): where the released canvas goes.
- PHASE 117 — the spool-effect latch (GP-IM-117): the release geometry at the line's end.
- PHASE 111 — the shuttle (GP-IM-111): the 35,000 kg airframe the gate protects.

§8 — DO-NOT-BUILD

- DO NOT exceed the gate — a line stronger than 25% of the bolt makes the airframe the fuse; that is the failure this phase bans.
- DO NOT mix the hierarchy — line < bracket < primary structure, consistently, with dynamic allowances.

- DO NOT skip the batch test — every line lot proves its snap point before it flies.
- DO NOT 'upgrade' to un-breaking anchors — rigid anchors are the snap-shock path back into the spine.

§9 — OWED

OWED: nothing — PASS with the recorded bolt calculation standing (the vet declined to reopen it). The 15 August blanket wording kill (the core objective's '100%') is already carried inline in §1.

§10 — STATUS / PROVENANCE / CHANGE CONTROL

STATUS: MANUAL GP-IM-115, v1.0 — CURRENT, seated law, master v2.1 (numerical print order). The one hundred and sixteenth manual of the program — 116 of 290. Built 23 August 2026, eighth of the 20-manual 'ok next 20' shift (the author's order, 23 August 2026: 'ok next 20'). First version until publication — 'until i publish it, it is still first version, otherwise is just typing' (his law, 22 August 2026).

PROVENANCE — THE STANDING ORDERS, VERBATIM: the town-trip order (seated in sitting 519): 'ok i have to go to town, i have converted them to pdfs now and am uploading, do the next 10 phases and i will read them when i get back' — the order that opened the manual program; the follow-on orders (seated in sittings 537, 557, 561, 562, 564, 566 and 572): 'ok next 10' / 'ok next ten' / 'all good next 10' / '10 more before go to the old lady for dinner' / 'do another 20 while i away so we clear the 100 mark for the day when i get back' / 'ok next 20'. The phase's own chain, all seated in the master: the contents line and the gate/conduit/parameter bodies (as seated — the Certificate IV carpentry and heavy-transport rigging masthead, verbatim); the 12 August naming batch (formerly 'STRUCTURAL SHEAR-COMPLIANT TRIPLE-CHUTE MATRIX'); the 15 August blanket kills carried inline; the 18 July origin text (the 25% law and the full figure set — 255 kN, 60 kN, 20 kN, 35,000 kg — verbatim); the vet batch 17 grade (PASS; the recorded bolt calculation not reopened) in §5.

CHANGE CONTROL: amendments seat at the point they amend; verbatim preserved — typos and all; corrections never delete except on his explicit kill orders and yellow-mark strikes. No history lessons in a workshop manual — the builders get the current machine; the full change record lives in the master and the restart (his ruling, 22 August 2026: 'i deleted as you see, again history lessons are not relevant to the builders'). Next update triggers: new line or bolt stock entering the supply chain, fuse-test data, or his word, or his word.

END OF MANUAL — PHASE 115